

Implementing Capacitive-Touch Interfaces in Low-Cost LED Lighting Designs



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In recent months, there has been a move to integrate capacitive-touch and capacitive-proximity user interfaces (UIs) into lighting applications. The simplicity of the UI, the ability to use irregularly-shaped sensors and the ability to seal the UI is advantageous for creating pleasing and low-maintenance LED interfaces. Unfortunately, differences in technology and techniques between lighting and touch-sensing can create conflicting design goals, particularly in the cost-constrained arena of architectural lighting. Let us see how we can bridge this gap.

Getting started with capacitive touch

Let us start with a basic overview of capacitive touch. A capacitor is essentially two conductors separated by an insulator. Depending upon the type of insulator, the area of conductors and the distance between these, the capacitor will have more or less capacitance. The following equation shows the basic relationship between various factors. C represents capacitance of the capacitor and A represents area of the overlap between the two conductors, the two physical constants ϵ_0 and ϵ_r represent permittivity of free space (ϵ_0) and rela-

tive permittivity (ϵ_r) of the insulating material, and D represents distance between the conductors.

$$C = \frac{\epsilon_0 \epsilon_r A}{D}$$

While the space between the plates usually contributes majority of the capacitance, there are also electric field lines connecting opposite sides of the capacitor that contribute to the capacitance of the capacitor. See Fig. 1 for an example plot of the electric field lines for a typical two-plate capacitor. In a capacitive-touch system, it is the electric field lines that project out from the capacitive, rather than the lines

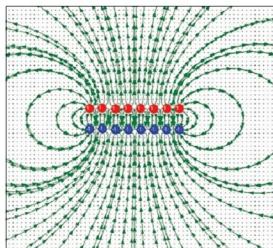


Fig. 1: An example plot of the electric field lines for a two-plate capacitor